

CS411 Lab 1 – Version 1

Jacob Thompson

Professor Thomas Kennedy

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Version 5

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1 Introduction

Algebra challenges many students, and traditional teaching methods worsen the problem. Classes often emphasize rote learning instead of providing meaningful context or engagement. This approach leaves students disinterested and frustrated. Recent data from the National Center for Education Statistics reveals just 26% of eighth-grade students across America achieved “math proficiency” in 2022—a decline that highlights serious concerns about current math education approaches.

Students struggle not just with algebra itself, but with how we teach it. They learn better when math feels like fun, not a chore. By making algebra interactive and personally rewarding, we can dramatically improve both their motivation and understanding. Any effective solution must engage students while making learning enjoyable.

Our solution, Rogue-Like Algebra, transforms algebraic concepts into engaging digital challenges, inviting learners to navigate a turn-based adventure where mathematical problem-solving becomes the key to progression. As players confront obstacles across various levels, they develop strategic thinking while remaining invested in their educational journey. By seamlessly integrating algebra into core gameplay mechanics, students continuously practice math without experiencing the tedium typically associated with traditional instruction.

The aim is simple: help students gain confidence in their math skills while keeping them engaged. By making learning feel less like a chore and more like an adventure, Rogue-Like Algebra hopes to offer a fresh approach to one of the most persistent problems in education.

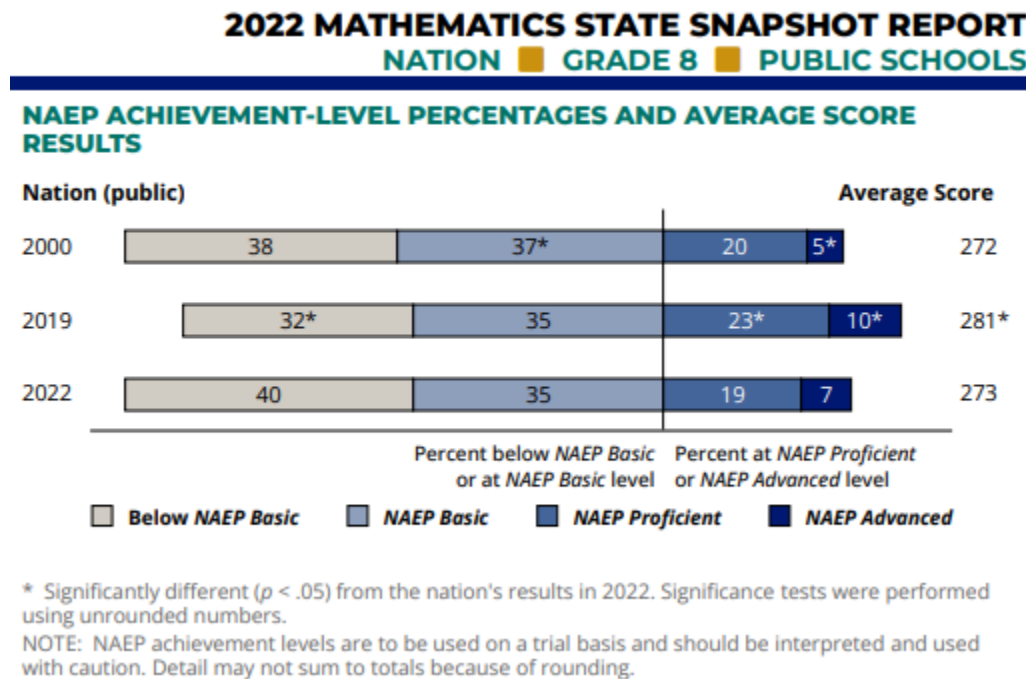


Figure 1: Graphic from the National Center for Education Statistics

2 Product Description

Rogue-Like Algebra is a turn-based adventure game that transforms the challenge of learning algebra into an engaging and immersive experience. Designed to address the struggles many face with algebraic concepts, the game blends strategic dungeon-crawling with problem-solving, where players use algebra to attack, defend, and progress. Mistakes carry consequences—such as reduced combat efficiency or character penalties.

The game's procedurally generated dungeons and adaptive difficulty ensure replay-ability and personalized challenge. Aimed primarily at middle and high school students but accessible to all learners, Rogue-Like Algebra provides an educational alternative to traditional rote learning by offering a gamified system that rewards creativity, risk-taking, and improvement. The game is available on the web and can be played in any modern browser. The project's goal is to foster confidence in math skills while making algebra fun, approachable, and rewarding.

2.1 Key Product Features and Capabilities

The key product features and capabilities of the game center around its engaging gameplay, which includes turn-based combat, an immersive storyline, and a variety of minigames designed to keep players interested and motivated. To maintain high levels of engagement, the game incorporates adaptive systems that scale difficulty based on player performance, introduces time limits to add a sense of urgency, and features challenging boss battles to test mastery. The primary objective is to gamify and simplify the process of learning algebra, making it more approachable and enjoyable for students.

This approach effectively addresses common challenges in learning algebra by rewarding players for solving difficult problems and completing in-game achievements, thereby reinforcing positive behavior and sustained effort. It also reduces the reliance on rote memorization by using procedural generation to create varied problems and allowing multiple accepted methods for solving them, which supports deeper understanding and flexibility in learning.

2.2 Major Components

The development of the game relies on several major components, carefully selected for their power, flexibility, and accessibility. The primary Integrated Development Environment (IDE) is Visual Studio Code, chosen for being powerful, highly extensible, and free to use. For version control, the team uses Git in combination with GitHub, which is not only the industry standard but also recommended by advisors, including a professor-investor. Continuous Integration and Continuous Deployment (CI/CD) are handled through GitHub Actions and Workflows, offering seamless integration with the project's existing GitHub repositories.

The game is targeted for deployment on both Windows and Linux operating systems. Windows remains the most widely used platform among end-users, while Linux serves as the foundational OS for many mobile devices, offering flexibility in distribution. Development is powered by the Godot Engine, a free and open-source platform known for its extensibility and community support and documentation. The core programming languages used are Rust and GDScript. Rust is selected for its speed, modern syntax, and strong safety guarantees, while GDScript is utilized for specific cases where Rust integration falls short or lacks engine-level support.

Several key systems are planned or in development to support the gameplay experience. Custom or adapted character controllers will be implemented using freely available templates. A leaderboard system will be developed to enhance competition and replayability. Procedural generation systems are essential and will be created for both dungeon layouts and dynamically generated math problems. Lastly, enemy AI must be designed and fine-tuned to offer a challenging yet fair experience to players.

3 Identification of Case Study

The primary target audience for the game consists of middle school and high school students, specifically those aged 12 to 18. These users are typically in the process of learning foundational algebra, making them an ideal demographic for an educational game that blends academic content with engaging gameplay. Additionally, the game is designed to appeal to STEM-oriented players and puzzle game enthusiasts, who are naturally drawn to logic-based challenges and problem-solving mechanics.

The secondary audience includes parents and educators, who often influence or make decisions about the educational tools and games students engage with. This group is likely to value the game's educational benefits and its ability to support learning in a fun, interactive way. Other secondary users include casual gamers and fans of indie games, as well as players with a general interest in logic and strategy-based games. These users may not be the core educational audience, but they can still find enjoyment and value in the game's mechanics and design.

The primary purpose of the game is to assist in the learning and reinforcement of foundational algebraic principles for the intended core audience—middle and high school students. The game is designed to deliver these concepts through methods that are both educationally effective and engaging. By integrating learning into an interactive format, it aims to maintain the interest and attention of students, offering a satisfying and enjoyable experience that supports the educational process.

In addition, however, it can help refresh and reinforce algebraic knowledge for users outside the core demographic, such as adults or older students who may benefit from a review of the material. Furthermore, the game is intended to provide a fun, repeatable gameplay loop that appeals to a broad audience, ensuring that players can continue to enjoy the game even beyond its educational value.

4 Glossary

4.1 Terms

Adaptive Learning – An educational method that adjusts content difficulty based on the learner's performance to personalize the learning experience.

Algebraic Concepts – Foundational mathematical principles involving variables, equations, expressions, and functions typically taught in middle and high school.

Continuous Integration (CI) – A development practice where developers frequently merge code changes into a shared repository, triggering automated builds and tests to detect issues early.

Continuous Deployment (CD) – A software release process that automatically delivers validated code changes to production, minimizing manual intervention and ensuring faster updates.

GDScript – A Python-like scripting language used in the Godot game engine, designed for creating game logic.

Git – A distributed version control system that tracks changes in source code during software development, enabling collaboration and history tracking.

GitHub – A cloud-based platform for hosting Git repositories. It offers tools for collaboration, issue tracking, code review, and CI/CD integration.

Godot – A free and open-source game engine used for developing 2D and 3D games, known for its flexibility and ease of use.

IDE (Integrated Development Environment) – A software suite that provides tools like a code editor, debugger, and compiler to streamline software development (e.g., VS Code).

Procedural Generation – A method of creating data algorithmically rather than manually, often used for generating levels, puzzles, or content dynamically in games.

Rogue-like – A subgenre of games typically featuring procedural generation, turn-based movement, and permanent death, emphasizing strategy and replayability.

Rust – A systems programming language known for performance, safety, and concurrency.

Turn-Based Combat – A gameplay mechanic where players and opponents take alternating turns to perform actions such as attacking or defending.

VS Code (Visual Studio Code) – A popular, lightweight code editor developed by Microsoft, used for writing and debugging code.

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